

tokens in which it is unclear whether the corresponding token truly encodes a PDC.

Error Analysis

Error analysis consisted of an array of data tabulation, visualization, and evaluation strategies; these are presented in the following subsections. Note that several additional methodologies were applied (e.g. error-to-feature ratio heat maps), but the results presented here were all produced using the methodologies specifically described in this section.

Classification Reports

Each language-classifier pair produced a classification report which included precision, recall, f-1 score, and support results for each construction used in the data set, as well as accuracy, macro average, and weighted average scores for each of these categories. These reports are useful for evaluating a given classifier's overall performance on classifying each construction in each language. The **precision** value is a fraction which represents the ratio of relevant retrieved instances (of a given construction) to total retrieved instances. The **recall** value is a fraction which represents the ratio of relevant retrieved instances to total relevant instances. The **f-1 score** is the harmonic mean of the precision and recall values. **Support** refers to the total number of instances of a given construction in the test data set.

Confusion Matrices

Each language-classifier pair also produced a confusion matrix, which tabulates in matrix form the relationship between actual constructions (rows) and predicted constructions (columns). The result is a matrix representation of results, both accurate and erroneous, for each language-classification pair. This matrix allows for visualizing the ratio of erroneous to accurate predictions made by the model; to these ratios a weighted score is calculated to account for frequency effects, such that, for example construction 1 may have a 100% error rate based on 1 representative token, whereas construction 2 may have a 50% error rate based on 75 representative tokens, suggesting that the construction 2 error rate is more informative than that for construction 1.

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7:36 PM Sep 15

Why are WS and WS range not discussed here?



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If you calculate these per category, how do you aggregate across categories to arrive at a score per language?



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What does this mean?



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Of what?



Error-Feature Reports

For each language-classifier pair, a report was constructed to tabulate the feature values of each specific error. In this report, each row represents an individual error, displaying the token number of the error, the predicted construction, and the actual construction; each column then shows the **feature values assigned to each construction**. From here, the error types can be sorted and grouped, providing a simple visualization of possible correlations between feature sets and error types. For example, in the **RFC** for Chinese, it was shown that all errors of predicted *le* → actual *suo...de* were resultatives and predominantly accomplishments, while all errors of predicted *le* → actual unmarked were resultatives and predominantly achievements.

Putting it All Together

These data were then tabulated and submitted to various **statistical analyses**, the results of which are presented in the following section, Results and Discussion. These tests included: X^2 values, X^2 patterns by way of p -values, ϕ -values, and weighted error rate scores. Using these values in tandem allows for error analysis of considerable richness, taking into account such useful metrics as error-to-feature ratio patterns, significance of features in error patterns, and strength of association between feature paradigms and error types. The results are present below.

Results and Discussion

Overview

Before diving into language-specific results, one generalization can be made regarding both the models' performance and also the methodology that produced the model inputs. The overall error rates by language were, perhaps unsurprisingly, quite different. The English data sets both contained all 150 tokens, and yielded overall error rates of 11/150 or 7.3% (RFC) and 13/150 or 8.7% (MLP), indicating relatively **strong** performance. German performed rather less well, with error rates of 36/149 or 24.2% (RFC) and 35/148 or 23.6% (MLP). Both models performed significantly worse on the Chinese data, yielding 52/115 or 45.2% (RFC) and 60/121 or 49.6% (MLP). There are a variety of reasons why this might be the case, but here I will highlight two specifically: anchor language effects and feature landscape effects.

Anchor Language Effects

Because English was used as the anchor language for this experiment, naturally every English token in the initial 300-token set was included, unlike with the German and Chinese which in some cases were deemed "non-equivalent". Even more significant, however, is the fact that the feature values for each token were assigned based on the English anchor text, meaning that the mapping between feature paradigm and **construction in English** is generally reliable; certainly there are ambiguous and other edge cases, but by and large the diagnostics used to evaluate feature values for each token are fairly well fleshed out and presumed to more or less accurately represent the features encoded by each token's PLC.



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But here you haven't discussed yet that you're using these composite constructions per language. That needs to be set up earlier too, when you discuss the constructions and the considerations that went into determining the category set for the classification experiment



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Report accuracy instead of error rate. Maybe later report error rate reduction. But typically we report accuracy (or better yet F1 scores)



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Marked as resolved



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Re-opened

Nevermind f1, accuracy is best here



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Compared to what?



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But at this stage I don't know what constructions English has. This has to be discussed (same for other two languages) earlier

